

Problem 6 - Stochastic Gradient Descent

i) $P(\text{not choosing a sample}) = \frac{n-1}{n} = 1 - \frac{1}{n}$

with replacement $\Rightarrow n$ independent events

$$\Rightarrow \left(1 - \frac{1}{n}\right) \times \dots \times \left(1 - \frac{1}{n}\right)$$

$$\Rightarrow \boxed{P_{\text{never}} = \left(1 - \frac{1}{n}\right)^n}$$

ii) $\lim_{x \rightarrow \infty} \left(1 + \frac{1}{x}\right)^x = e$; Show that $\lim_{n \rightarrow \infty} P_{\text{never}} \approx 0.3674$

$$\lim_{n \rightarrow \infty} \left(1 - \frac{1}{n}\right)^n$$

$$\text{let } y = \left(1 - \frac{1}{n}\right)^n$$

$$\ln(y) = n \ln\left(1 - \frac{1}{n}\right)$$

$$\lim_{n \rightarrow \infty} \ln(y) = \lim_{n \rightarrow \infty} \frac{\ln\left(1 - \frac{1}{n}\right)}{\frac{1}{n}} = \frac{\left(\frac{-1}{1-\frac{1}{n}}\right)\left(\frac{1}{n^2}\right)}{-\frac{1}{n^2}} = -1$$

$$\Rightarrow \lim_{n \rightarrow \infty} \ln(y) = -1$$

$$\Rightarrow \boxed{\lim_{n \rightarrow \infty} y = e^{-1} = 1/e \approx 0.3679}$$

iii) One random observation $\text{Var}(\nabla f(x^{(i)}, y^{(i)})) = \sigma^2$
Let B be the batch size of a mini batch. Given B independent random observations, $(x_1, \dots, x_B)$

$$\text{mean } \bar{X} = \frac{1}{B} \sum_{i=1}^B x_i \quad \text{where } \text{Var}(x_i) = \sigma^2$$

$$\text{Var}(\bar{X}) = \text{Var}\left(\frac{1}{B} \sum_{i=1}^B x_i\right) = \left(\frac{1}{B}\right)^2 \text{Var}\left(\sum_{i=1}^B x_i\right) = \frac{1}{B^2} (B\sigma^2) = \boxed{\frac{\sigma^2}{B}}$$

iv) Scaling mini batch of size B by k

$$\Rightarrow \text{Var}(\bar{X}) = \text{Var}\left(\frac{1}{kB} \sum_{i=1}^{kB} x_i\right) = \frac{1}{k^2 B^2} kB\sigma^2 = \boxed{\frac{1}{kB} \sigma^2}$$

v) The claim in the paper is wrong because increasing the batch size and learning rate by a factor of k increases the variance.

Let the variance of the gradient of one observation

$$\text{Var}(\nabla f(x^{(i)}, y^{(i)})) = \sigma^2$$

The variance of a mini batch of size B ,

$$\text{Var}\left(\frac{1}{B} \sum_{i=1}^B x_i\right) = \frac{\sigma^2}{B} \text{ without changing the learning rate}$$

Scaling the mini batch size by k gives the following

$$\text{Var}\left(\frac{1}{kB} \sum_{i=1}^{kB} x_i\right) = \frac{\sigma^2}{kB} \text{ without changing the learning rate}$$

Scaling the learning rate by k , the variance of a

mini batch of size kB gives

$$\text{Var}\left(\frac{k}{kB} \sum_{i=1}^{kB} x_i\right) = \frac{k^2 \sigma^2}{kB} = \frac{k \sigma^2}{B}, \text{ thus increasing the variance}$$

To maintain the variance of $\frac{\sigma^2}{B}$, one should scale the batch size by k and scale the learning rate

by $\sqrt{k}$.

$$\Rightarrow \text{Var}\left(\frac{\sqrt{k}}{kB} \sum_{i=1}^{kB} x_i\right) = \frac{k}{k^2 B^2} kB \sigma^2 = \boxed{\frac{\sigma^2}{B}}$$